Hybrid LSTM-IoT in Agriculture: A Systematic Literature Review

1st Eddy Nurraharjo
Dept. of Information Tech. and Industry
Universitas Stikubank
Semarang, Indonesia
eddynurraharjo@edu.unisbank.ac.id
ORCID: 0000-0003-2015-5993

2nd Ema Utami
Dept. of Informatics Doctorate
Universitas Amikom
Yogyakarta, Indonesia
ema.u@amikom.ac.id
ORCID: 0000-0002-8237-8693

3rd Kusrini
Dept. of Informatics Doctorate
Universitas Amikom
Yogyakarta, Indonesia
kusrini@amikom.ac.id
ORCID: 0000-0001-9573-3909

4th Kumara Ari Yuana
Dept. of Informatics Doctorate
Universitas Amikom
Yogyakarta, Indonesia
kumara.a@amikom.ac.id
ORCID: 0000-0003-4864-8693

2024 International Conference on Information Technology and Computing (ICITCOM) | 979-8-3503-7983-9/24/$31.00 ©2024 IEEE | DOI: 10.1109/ICITCOM62788.2024.10762212

***Abstract*—Climate conditions significantly influence agricultural productivity, with regional fluctuations impacting both crop yield and quality. This study explores the significant impact of climate conditions on agricultural productivity and how innovative solutions like controlled environments are being adopted to mitigate these effects. It systematically reviews the integration of Hybrid LSTM (Long Short-Term Memory) and IoT (Internet of Things) technologies in agriculture, focusing on their applications, recent advancements, and challenges. We examined 49 articles of Scopus. The research paper concludes with a compelling overview of agricultural themes frequently explored: irrigation and soil dominate at approximately 17% of the analyzed literature each. Equally significant are topics such as rainfall patterns, climate dynamics, network management, remote sensing applications, greenhouse control mechanisms, and evapotranspiration, each capturing around 38.4% of scholarly focus. Additionally, farming practices, crop management strategies, and disease control strategies collectively account for about 19.1% of the studies reviewed. Less frequent yet pertinent themes like robotics in agriculture, efficient water usage, and advanced pest detection methods are each highlighted at approximately 12.8%. Similarly, topics like production economics, framework development, UAV systems, hydroponics, plantation management, and flood mitigation strategies also command approximately 12.8% of scholarly attention. This comprehensive analysis underscores the diverse array of interests within agricultural research, reflecting the necessity of a multifaceted approach to address the complex challenges in the field.**

Keywords—LSTM, IoT, Hybrid, Deep Learning, Review

I. INTRODUCTION

In the present era, the combination of Internet of Things (IoT) technology with Hybrid LSTM approaches has become a central focus in agricultural research. "Hybrid LSTM-IoT in Agriculture: A Systematic Literature Review" provides a thorough examination of this partnership's importance in digitally transforming the agricultural sector[1], [2]. Recent studies underscore that this integration not only holds potential for improving production efficiency and resource management but also offers innovative solutions to address the complex challenges faced by farmers today[3], [4]. By merging artificial intelligence with advanced IoT capabilities, this approach facilitates more accurate predictions of crop conditions and weather patterns, thereby establishing a strong foundation for sustainable agriculture in the future.

As the global population continues to grow[5], the urgency of meeting the needs of this expanding demographic becomes more pronounced. The demand for data and information, whether continuous or real-time, is expected to rise accordingly. The integration of Information and Communication Technology (ICT) systems plays a pivotal role in meeting this growing demand[6]. This article explores one such integration: LSTM, renowned for its ability to capture temporal data relationships[7], combined with IoT[8], which enables real-time data streams from diverse sensors[9]. The adoption of these technologies holds significant promise for enhancing operational efficiency and productivity across agricultural sectors[10], [11], [12].

This systematic literature review aims to comprehensively evaluate the application of LSTM, IoT, and Hybrid LSTM-IoT in agricultural contexts. By synthesizing current research findings, the review not only provides an overview of advancements in this field but also identifies existing knowledge gaps. These gaps present opportunities for further research and development, crucial for advancing the effectiveness and scalability of integrated technologies in agriculture[13]. Furthermore, the review seeks to offer insights and guidance to researchers, policymakers, and agricultural stakeholders looking to deepen their understanding and leverage these innovative integrations to enhance agricultural outcomes globally.

II. METODHOLOGY

A. Review Protocol

The Systematic Literature Review (SLR) process is divided into three stages: planning, execution, and reporting. During the planning stage, particular emphasis was placed on formulating pertinent research questions, identifying appropriate keywords, and selecting suitable publication databases. A variety of digital library resources will be meticulously selected in order to ensure comprehensive coverage of the relevant topics. In the execution phase, a comprehensive search is conducted across the various established databases, followed by the application of rigorous selection criteria to identify relevant studies. Subsequently, the selected papers are collated, and pertinent data is extracted in order to respond to the research questions. In the reporting stage, the collected data was synthesized in order to answer the research questions. The results were then presented using graphs and summary tables.

Authorized licensed use limited to: UNIVERSITAS GADJAH MADA. Downloaded on January 22,2025 at 06:13:41 UTC from IEEE Xplore. Restrictions apply.

B. Research Question

The primary objective of this article is to examine international publications that focus on research themes employing LSTM, IoT, and their hybrid or ensemble approaches. The framework utilized to construct the research questions is based on the PICOC methodology (Population, Intervention, Comparison, Outcomes, and Context), as outlined in Table 1.

TABLE I. FRAMEWORK

Population	Agriculture
Intervention	Use of Hybrid LSTM-IoT
Comparison	-
Outcome	Model performance, classification, comparation, and improvements
Context	Irrigation, cultivation, climate, weather

Using this PICOC framework, we have formulated the research questions and motivations, as detailed in Table 2.

TABLE II. RESEARCH QUESTION

RQ	Question	Motivation
1	What is the current state of research on LSTM applications in agriculture?	This will aid in comprehending the current literature regarding the application of LSTM in agriculture.
2	What are the existing IoT implementations in agriculture?	This will allow for a thorough understanding of how IoT has been utilized within agricultural practices.
3	How have LSTM and IoT been combined in agricultural settings?	This inquiry will assist in exploring studies that detail the merging of LSTM and IoT in agricultural environments.
4	What are the reported benefits and challenges of utilizing a hybrid LSTM-IoT approach in agriculture?	This will facilitate the evaluation of the advantages and obstacles presented by the hybrid LSTM-IoT approach in agriculture.

C. Search Query Approach

This comprehensive approach is essential for achieving broad literature coverage. The digital databases that were searched include Google Scholar (scholar.google.com), IEEE-Xplore (ieeexplore.ieee.org), ScienceDirect (sciencedirect.com), and Scopus (scopus.com). The identification of search terms was conducted in accordance with the PICOC framework, research questions, relevant titles, abstracts, and keywords.

TABLE III. SEARCH QUERY

RQ	Context			
	Search query	***Find paper***	***Select. paper***	***Scr. paper***
1	(lstm* OR long* OR long-*) AND agricul* AND climat* AND predict*	2827	39	12
2	(iot* OR internet AND of* OR -iot*) AND agricul* AND climat* AND predict*	357	28	12
3	(lstm* OR long* OR long-* OR *short*) AND (iot* OR internet-of* OR -iot*) AND irrigat*	368	25	13
4	(challe* OR benef*) AND (lstm* OR long* OR long-* OR *short*) AND (iot* OR internet-of* OR -iot*) AND agric*	429	39	16

Subsequently, the search keywords were tailored to meet the specific requirements of each database utilized. The searches were conducted by considering the keywords, titles, and abstracts of relevant studies. The results are summarized in Table 3.

The search period was limited to the years 2022 to 2024 to ensure the currency of the information, Figure 1.

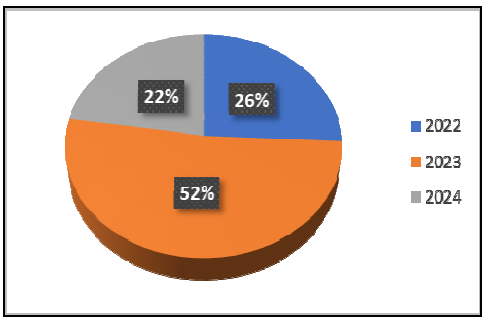


Fig. 1. Papers distribution by year

Furthermore, we limited the publication types to journal articles and included only those articles available in English. This approach ensures that our search is focused on high-quality and relevant studies that align with the scope of our literature review. The search results, based on the selected keywords, yielded information on the distribution year classification that was deemed pertinent and suitable for the purposes of this article. The quality of the review paper is classified according to the quartile scale of the Scopus indexation. As illustrated by the distribution of papers in Figure 2.

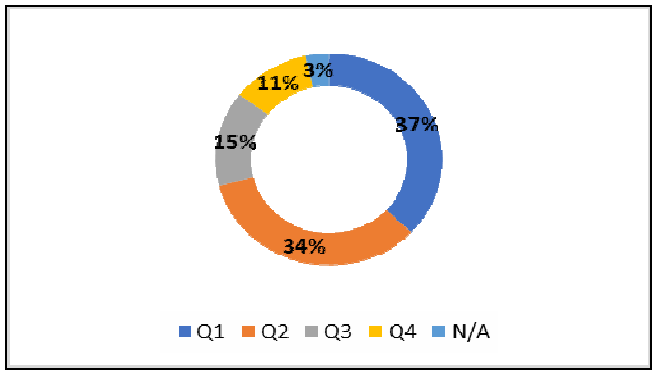


Fig. 2. Papers distribution by quartile

This categorization is based on the calculation of the number of citations of journals in the Scopus database. The journals with the highest number of citations will be placed in the first quartile (Q1), while those with the lowest number of citations will be placed in the fourth quartile (Q4). The quartile of a journal can be determined through the official Scopus website or Scimago Journal & Country Rank (SJR).

D. Selection Study

These studies concentrate on the field of computer science, particularly in the areas of cultivation, agriculture, and climate. They utilise datasets of varying scales and discuss and compare modelling performance in the context of LSTM, IoT, and hybridised approaches. Only the final, published version of a study will be included, and only the most recent, complete journal version will be selected.

Authorized licensed use limited to: UNIVERSITAS GADJAH MADA. Downloaded on January 22,2025 at 06:13:41 UTC from IEEE Xplore. Restrictions apply.

Conversely, exclusions encompass studies that do not align with the context of LSTM, IoT, and the hybridization of the two approaches, as well as studies that do not pertain to the agricultural computer science environment. Additionally, studies that are not written in English will be excluded. The organization and management of search outcomes was conducted using the Mendeley Reference Manager software.

E. Data Extraction

The data from the selected primary studies were extracted in order to contribute to the answering of the relevant research questions in this review. For each of the 49 selected primary studies, data extraction table fields were populated in Microsoft Excel format. The table fields were designed with the specific intention of collecting the necessary data from the primary studies in order to answer the research questions. Based on the research question and the analysis we wished to introduce, four criteria were identified that were used to answer the research question, as outlined in Table 4.

III. RESULT

The study quality assessment can be employed to inform the interpretation of the synthesis findings and to define the strength of the inferences that have been elaborated. The objective of data synthesis is to integrate evidence from the selected studies in order to address the research questions. The strength of a single piece of evidence may be relatively weak, but the aggregation of multiple pieces of evidence can enhance the overall strength of the argument. The data extracted in this review encompass both quantitative and qualitative data.

TABLE IV. CITATION RESULT

Result	Citation papers
Research on LSTM	[14], [15], [16], [17], [18], [19]
Research on IoT	[20], [21], [22], [23], [24], [25]
Research on LSTM-IoT combination	[23], [26], [27], [28], [29], [30]
Challenges on Research	[31], [32], [33], [34]

A variety of techniques were employed to synthesize the extracted data pertaining to different types of research questions. In general, the narrative synthesis method was employed. The data were organized in a manner consistent with the research questions. The selected papers have been classified into five groups based on the applied methods used in their respective studies, Figure 3.

The chart illustrates the potential review of various methods, including IoT, LSTM, hybrid, combined, and other approaches, from the perspective of current research. In contemporary studies, IoT (Internet of Things) stands out with a score of around 20, highlighting its importance as a foundational technology in the development of real-time connected automation and control systems. IoT continues to evolve, providing efficient and effective solutions across industries by enabling wide-scale device connectivity.

IoT: The Internet of Things is a rapidly evolving field of study that encompasses a diverse range of disciplines[35], [36]. This category encompasses research studies in computer science applied to agriculture that predominantly employ IoT-based devices with various programs and algorithms to address their research problems[37], [38].

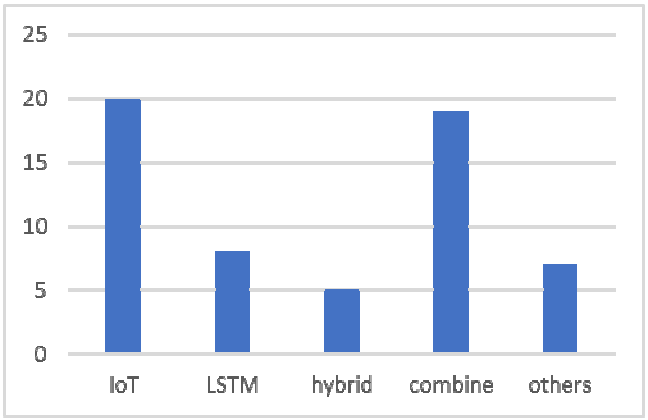


Fig. 3. Method study

LSTM: This classification encompasses research studies in computer science applied to agriculture that employ the LSTM method in an effective manner to address their research issues[39].

Hybrid: This category encompasses research studies in computer science applied to agriculture that integrate two or more distinct methods or algorithms to create a novel, more efficacious approach[40], [41].

Combine: This classification encompasses research studies in computer science applied to agriculture that employ a combination of multiple methods or algorithms to address distinct aspects of a problem or achieve a comprehensive solution. These methods can be employed independently or in conjunction with one another, thereby providing a complementary approach to problem-solving[40].

Others: This category encompasses research studies in computer science applied to agriculture that employ alternative methodologies[11], [42], [43].

TABLE V. HYBRID MODEL UTILIZATION

Hybrid Model	Research Study	Comparison model
CNN-BiLSTM	rainfall prediction[44]	Bi-LSTM
LSTM	climate change prediction	Stack-LSTM, Bi-LSTM, CNN-LSTM
LSTM-PSO	climate forecasting[45]	Simple-RNN, LSTM, LSTM-GRU
EDT-LSTM	soil moisture prediction[46]	Encoder-Decoder LSTM, LSTM
LSTM-RNN	soil moisture prediction[39]	MLP, LR
ConvLSTM2D	plant genotypes grown[47]	FCN, Conv2D, Encoder, LSTM, ResNet
LSTM	forecast crop yield[18]	RNN
LSTM-PCA	rainfall prediction[48]	KNN, LR, SVM, RF, NB, ANN
stacked-LSTM	feature selection[16]	DCT, KNN, LR
LSTM-Adaboost	Sea surface temperature prediction[49]	LSTM, AdaBoost, SVR, BPNN, S_LSTM-AdaBoost
ensemble LSTM	evapotranspiration prediction[50]	LSTM

Table 5 provides a comprehensive review of hybrid models utilized in various research studies, emphasizing their effectiveness in specialized prediction and forecasting tasks across different fields. Several papers have highlighted

Authorized licensed use limited to: UNIVERSITAS GADJAH MADA. Downloaded on January 22,2025 at 06:13:41 UTC from IEEE Xplore. Restrictions apply.

intriguing comparison methods, particularly in comparing hybrid LSTM models with other combinations of LSTM models. These comparisons consistently demonstrate the superior performance of hybrid LSTM models over alternative methods. This trend is evident across multiple findings listed in this table, underscoring the significant advantages of hybrid approaches in enhancing predictive accuracy and reliability in diverse scientific applications.

IV. Discussion

Advances in LSTM. Based on the research questions raised in this systematic study, several important aspects are worth discussing. Multiple studies focus on various applications of LSTM and other deep learning models in predictions related to agriculture and climate. For example, the EDT-LSTM model outperforms traditional LSTM in soil moisture prediction. LSTM also shows high accuracy in rainfall prediction systems, addressing complexity issues effectively. The hybrid MEA-DBN-LSTM model provides more accurate solar radiation predictions compared to other empirical models, while the CFS-based regression model and LSTM algorithm succeed in accurate crop selection. Additionally, using LSTM with the BER optimization algorithm improves rainfall prediction accuracy and aids in better water management. In agricultural commodity price prediction, LSTM demonstrates highly accurate results. The LSTM-M-PSO model is also effective in enhancing rainfall prediction accuracy, tackling climate variation challenges. This field of study finds that time-series-based LSTM significantly boosts climate parameter prediction accuracy. The m-LSTM model is used to evaluate the impact of climate change on rice production, providing sustainable guidance for rice farming. Finally, machine learning techniques with Logistic Regression and LSTM enhance the accuracy of plant disease detection and soil content prediction, offering promising solutions to agricultural challenges. Overall, this research highlights that LSTM and other deep learning models play a crucial role in improving prediction accuracy across various aspects of agriculture and climate, supporting better and more sustainable decision-making.

Advances in IoT. The utilization of IoT technology in agriculture has significantly boosted productivity by addressing challenges such as soil fertility decline and intensive pest attacks. Climate change affects agricultural productivity, necessitating early detection for effective pest management strategies. The integration of AI and IoT is crucial for predicting plant diseases and enhancing agrotechnical monitoring systems. Agriculture 4.0 optimizes farming practices with efficient resource allocation and adaptation to climate change. The development of cost-effective IoT agricultural networks has been effective in addressing global food insecurity. IoT technology and machine learning aid in predicting harvest outcomes and farmers' decision-making processes. AI/ML models are essential in transforming data into useful information, although obtaining high-quality data remains a challenge. With the projected increase in the world's population, updating agricultural systems using Smart Farm IoT is crucial. Automatic data collection by IoT sensors enables accurate harvest predictions. The escalating demand for food exacerbates global plant diseases, with AI offering effective solutions in detecting and combating plant diseases.

Advances in Hybrid Systems. IoT technology is a crucial asset in enhancing agricultural productivity by utilizing predictive pest attacks based on environmental parameters, alongside soil moisture models providing vital information for sustainable irrigation. IoT-based irrigation networks emerge as promising solutions to address global water shortages by optimizing water usage, while IoT-based Evapotranspiration (ET) estimates offer efficient alternatives in reducing soil salinity. Automated irrigation systems, powered by IoT technology, help reduce water wastage and enhance agricultural efficiency, alongside the role of Agriculture 4.0 introducing precision irrigation to support sustainable farming. The combination of sensor technology and IoT is key in supporting location-based decision-making in precision farming, while IoT-based algorithms optimize groundwater management with more cost-effective solutions. On the other hand, IoT networks deployed in urban and rural areas improve data transmission in agriculture, while Deep Learning and IoT collaborate to enhance irrigation efficiency and water usage. Smart farming approaches with IoT result in more efficient resource utilization in irrigation, while AI-based applications enable automation and optimization of irrigation processes in precision agriculture. Support for adaptive agricultural models with IoT brings increased crop yields and resource efficiency, while IoT-enabled smart greenhouses help regulate plant growth conditions for optimal yields.

Challenges. In the prediction of increasing urbanization by 2050, pressure on resources such as food and freshwater are becoming more pronounced, with the potential for water shortages looming. Urban agriculture and Precision Agriculture, supported by IoT and ML technologies, offer solutions for urban food security and sustainability. Through in-depth analysis, this article proposes a smart irrigation system based on soil and weather conditions, providing a cost-effective solution for local weather monitoring, while Accurate Evapotranspiration (ET) estimates are essential for acidic soils, and IoT and machine learning-based models show potential to enhance accuracy. It underscores the crucial role of agriculture in addressing global challenges like food security and climate change, supported by Agriculture 4.0 and IoT. This paper introduces a sophisticated IoT-based system for precision crop irrigation, using sensors to monitor various environmental factors and enabling real-time data utilization to minimize errors and enhance predictions, empowering farmers to make informed irrigation decisions. Precision Agriculture (PA) leverages sensor technology and DCNN for real-time decision support as well as the introduction of AIDSII, an AI-based digital application integrating IoT-based precision farming and CNN-LSTM models to provide comprehensive feedback to farmers for automated and optimized irrigation processes, with advanced IoT solutions for coastal agriculture and aquaculture regions addressing emerging challenges. LPWAN, with low power consumption, affordable cost, and extensive geographic coverage, becomes a prime choice among researchers, and the development of Wireless Underground Sensor Networks (WUSNs) gains attention for year-round subsurface data collection without disrupting surface activities. With increasing global water supply demand, water conservation technology in agriculture becomes increasingly critical, with Information and Communication Technology (ICT), including soil sensors, holding potential to enhance crop productivity and quality

Authorized licensed use limited to: UNIVERSITAS GADJAH MADA. Downloaded on January 22,2025 at 06:13:41 UTC from IEEE Xplore. Restrictions apply.

with minimal resource usage and environmental impact, along with the integration of Internet of Things (IoT) technology in agriculture emerging as a transformative force.

The accompanying figure 4, illustrates the scientific concentrations obtained from the extraction of the selected paper themes, distributed across the main study themes. The most frequently addressed topics are irrigation and soil, each appearing in approximately 17% of the analyzed papers. Other significant themes include rainfall, climate, network management, remote sensing, greenhouse control, and evapotranspiration, each representing around 38,4% of the focus. Themes such as farming practices, crop management, and disease control are highlighted in about 19,1% of the studies. Less frequently mentioned but still relevant themes, like robotics, water usage, pest detection each representing around 12,8% and production prices, framework development, UAV systems, hydroponics, plantations, and flood management, each account for around 12,8% of the concentration. This analysis underscores the diverse areas of focus in agricultural research, reflecting the multifaceted approach required to address the complex challenges in the field.

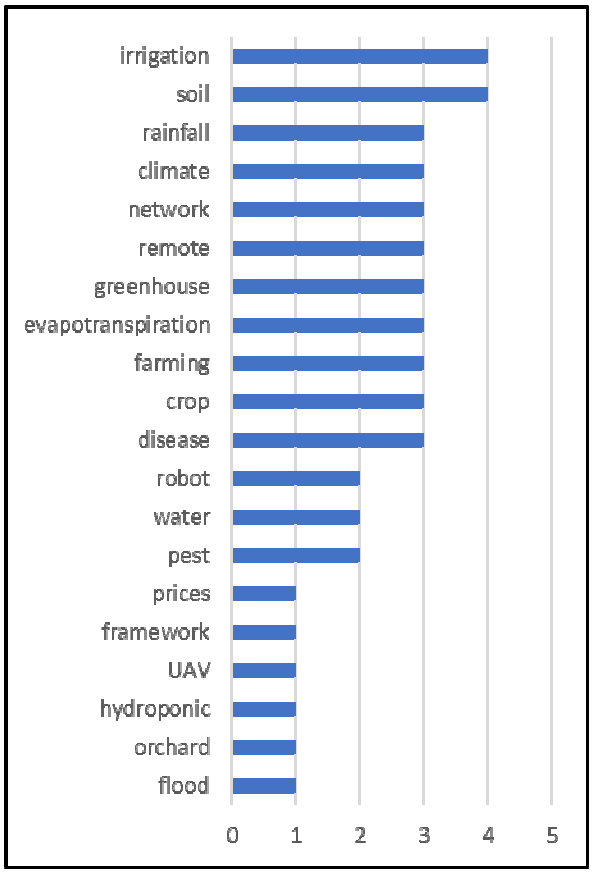


Fig. 4. Concern study

V. CONCLUSION

This systematic literature review elucidates the multifaceted impact of Hybrid LSTM and IoT technologies on agriculture, emphasizing their role in optimizing productivity and sustainability. The integration of these technologies addresses crucial agricultural parameters such as irrigation, soil quality, rainfall, and climate conditions. The study explores the advancements in IoT sensors and communication systems, highlighting their application in greenhouse management, remote monitoring, and water regulation. Moreover, it examines the efficacy of hybrid approaches in combating challenges like disease and pest control, while leveraging robotics for enhanced farming practices. By synthesizing these insights, the research provides a comprehensive guide for farmers, offering effective, cost-efficient, and durable solutions to modern agricultural challenges, thus paving the way for innovative and sustainable agricultural practices.

REFERENCES

[1] A. Simo, S. Dzitac, A. Duțu, and I. Pandelica, "Smart Agriculture in the Digital Age: A Comprehensive IoT-Driven Greenhouse Monitoring System," *International Journal of Computers, Communications and Control*, vol. 18, no. 6, 2023, doi: 10.15837/ijccc.2023.6.6147.

[2] N. C. Sarkar *et al.*, "Enhancing livelihoods in farming communities through super-resolution agromet advisories using advanced digital agriculture technologies," *Journal of Agrometeorology*, vol. 25, no. 1, pp. 68–78, 2023, doi: 10.54386/jam.v25i1.2080.

[3] R. Benameur, A. Dahane, B. Kechar, and A. E. H. Benyamina, "An Innovative Smart and Sustainable Low-Cost Irrigation System for Anomaly Detection Using Deep Learning," *Sensors*, vol. 24, no. 4, 2024, doi: 10.3390/s24041162.

[4] E. Jacques, A. Neuenfeldt Júnior, S. De Paris, M. Francescatto, and J. Siluk, "Smart cities and innovative urban management: Perspectives of integrated technological solutions in urban environments," *Heliyon*, vol. 10, no. 6, p. e27850, 2024, doi: https://doi.org/10.1016/j.heliyon.2024.e27850.

[5] H. Chen, J. Sun, and H. Li, "Increased population exposure to precipitation extremes under future warmer climates," *Environmental Research Letters*, vol. 15, no. 3, 2020, doi: 10.1088/1748-9326/ab751f.

[6] L. Charfeddine and M. Umlai, "ICT sector, digitization and environmental sustainability: A systematic review of the literature from 2000 to 2022," *Renewable and Sustainable Energy Reviews*, vol. 184, p. 113482, 2023, doi: https://doi.org/10.1016/j.rser.2023.113482.

[7] Kurnianingsih *et al.*, "Big data analytics for relative humidity time series forecasting based on the LSTM network and ELM," *International Journal of Advances in Intelligent Informatics*, vol. 9, no. 3, pp. 537–550, 2023, doi: 10.26555/ijain.v9i3.905.

[8] Mahaveerakannan R, C. Anitha, Aby K Thomas, S. Rajan, T. Muthukumar, and G. Govinda Rajulu, "An IoT based forest fire detection system using integration of cat swarm with LSTM model," *Comput Commun*, vol. 211, pp. 37–45, 2023, doi: 10.1016/j.comcom.2023.08.020.

[9] G.-X. Liu, L.-F. Shi, and D.-J. Xin, "Data Integrity Monitoring Method of Digital Sensors for Internet-of-Things Applications," *IEEE Internet Things J*, vol. 7, no. 5, pp. 4575–4584, 2020, doi: 10.1109/JIOT.2020.2967504.

[10] R. Lassoued, P. W. B. Phillips, and S. J. Smyth, "Exploratory analysis on drivers and barriers to Canadian prairie agricultural technology innovation and adoption," *Smart Agricultural Technology*, vol. 5, p. 100257, 2023, doi: https://doi.org/10.1016/j.atech.2023.100257.

[11] J. H. Al-Ammary and M. E. Ghanem, "Information and communication technology in agriculture: awareness, readiness and adoption in the Kingdom of Bahrain," *Arab Gulf Journal of Scientific Research*, vol. 42, no. 1, pp. 182–197, 2024, doi: 10.1108/AGJSR-07-2022-0113.

[12] G. Langer, H. Schulze, and S. Kühl, "From intentions to adoption: Investigating the attitudinal and emotional factors that drive IoT sensor use among dairy farmers," *Smart Agricultural Technology*, vol. 7, 2024, doi: 10.1016/j.atech.2024.100404.

[13] R. Rani, J. Sahoo, S. Bellamkonda, S. Kumar, and S. K. Pippal, "Role of Artificial Intelligence in Agriculture: An Analysis and Advancements With Focus on Plant Diseases," *IEEE Access*, vol. 11, pp. 137999–138019, 2023, doi: 10.1109/ACCESS.2023.3339375.

[14] P. J. Vaz, G. Schutz, C. Guerrero, and P. J. S. Cardoso, "Hybrid Neural Network Based Models for Evapotranspiration Prediction Over Limited Weather Parameters," *IEEE Access*, vol. 11, pp. 963–976, 2023, doi: 10.1109/ACCESS.2022.3233301.

[15] L. Xing *et al.*, "Predicting daily solar radiation using a novel hybrid long short-term memory network across four climate regions of China," *Comput Electron Agric*, vol. 212, 2023, doi: 10.1016/j.compag.2023.108139.

Authorized licensed use limited to: UNIVERSITAS GADJAH MADA. Downloaded on January 22,2025 at 06:13:41 UTC from IEEE Xplore. Restrictions apply.

[16] I. Shanmugam, J. Rethnaraj, and G. Mani, "Selection of crop varieties and yield prediction based on phenotype applying deep learning," *International Journal of Electrical and Computer Engineering*, vol. 13, no. 6, pp. 6806–6816, 2023, doi: 10.11591/ijece.v13i6.pp6806-6816.

[17] S. Thirumal and R. Latha, "Machine Learning based Predictive Assessments of Impacts of Influential Climatic Conditions for the Sustainable Productivity of Paddy Crops," *SSRG International Journal of Electronics and Communication Engineering*, vol. 10, no. 5, pp. 116–127, 2023, doi: 10.14445/23488549/IJECE-V10I5P111.

[18] I. Shanmugam, J. Rethnaraj, S. Rajendran, and S. Manickam, "Prediction on field crops yield based on analysis of deep learning model," *Indonesian Journal of Electrical Engineering and Computer Science*, vol. 30, no. 1, pp. 518–527, 2023, doi: 10.11591/ijeecs.v30.i1.pp518-527.

[19] E.-S. M. El-Kenawy, A. A. Abdelhamid, F. Alrowais, M. Abotaleb, A. Ibrahim, and D. S. Khafaga, "Al-Biruni Based Optimization of Rainfall Forecasting in Ethiopia," *Computer Systems Science and Engineering*, vol. 45, no. 3, pp. 2885–2898, 2023, doi: 10.32604/csse.2023.034206.

[20] W. Cai, R. Wei, L. Xu, and X. Ding, "A method for modelling greenhouse temperature using gradient boost decision tree," *Information Processing in Agriculture*, vol. 9, no. 3, pp. 343–354, 2022, doi: 10.1016/j.inpa.2021.08.004.

[21] F. M. Talaat, "Crop yield prediction algorithm (CYPA) in precision agriculture based on IoT techniques and climate changes," *Neural Comput Appl*, vol. 35, no. 23, pp. 17281–17292, 2023, doi: 10.1007/s00521-023-08619-5.

[22] R. N. Bashir *et al.*, "Smart reference evapotranspiration using Internet of Things and hybrid ensemble machine learning approach," *Internet of Things (Netherlands)*, vol. 24, 2023, doi: 10.1016/j.iot.2023.100962.

[23] R. M. Saleem *et al.*, "Internet of Things Based Weekly Crop Pest Prediction by Using Deep Neural Network," *IEEE Access*, vol. 11, pp. 85900–85913, 2023, doi: 10.1109/ACCESS.2023.3301504.

[24] R. M. Nadeem, A. Jaffar, and R. M. Saleem, "IoT and machine learning based stem borer pest prediction," *Intelligent Automation and Soft Computing*, vol. 31, no. 3, pp. 1377–1392, 2022, doi: 10.32604/IASC.2022.020680.

[25] I. Laktionov, G. Diachenko, D. Rutkowska, and M. Kisiel-Dorohinicki, "AN EXPLAINABLE AI APPROACH TO AGROTECHNICAL MONITORING AND CROP DISEASES PREDICTION IN DNIPRO REGION OF UKRAINE," *Journal of Artificial Intelligence and Soft Computing Research*, vol. 13, no. 4, pp. 247–272, 2023, doi: 10.2478/jaiscr-2023-0018.

[26] V. Y. Chandrappa, B. Ray, N. Ashwatha, and P. Shrestha, "Spatiotemporal modeling to predict soil moisture for sustainable smart irrigation," *Internet of Things (Netherlands)*, vol. 21, 2023, doi: 10.1016/j.iot.2022.100671.

[27] A. Tschand, "Semi-supervised machine learning analysis of crop color for autonomous irrigation," *Smart Agricultural Technology*, vol. 3, 2023, doi: 10.1016/j.atech.2022.100116.

[28] A. Benbatouche and B. Kadri, "Design and realization of low-cost solenoid valve remotely controlled, application in irrigation network," *Bulletin of Electrical Engineering and Informatics*, vol. 11, no. 3, pp. 1779–1788, 2022, doi: 10.11591/eei.v11i3.4123.

[29] G. Routis and I. Roussaki, "Low Power IoT Electronics in Precision Irrigation," *Smart Agricultural Technology*, vol. 5, 2023, doi: 10.1016/j.atech.2023.100310.

[30] L. Liu *et al.*, "An Edge-computing flow meter reading recognition algorithm optimized for agricultural IoT network," *Smart Agricultural Technology*, vol. 5, 2023, doi: 10.1016/j.atech.2023.100236.

[31] E. M. Raouhi, M. Zouizza, M. Lachgar, Y. Zouani, H. Hrimech, and A. Kartit, "AIDSII: An AI-based digital system for intelligent irrigation," *Software Impacts*, vol. 17, 2023, doi: 10.1016/j.simpa.2023.100574.

[32] W.-T. Sung, C.-H. Weng, and S.-J. Hsiao, "Smart Greenhouse Control via NB-IoT," *Intelligent Automation and Soft Computing*, vol. 34, no. 3, pp. 1971–1988, 2022, doi: 10.32604/iasc.2022.026927.

[33] N. T. E. Figueroa, V. A. Priya, S. K. Shanmugam, K. V. Kumar, S. Sengan, and A. M. C. Bolivar, "Adaptive Approach to Anomaly Detection in Internet of Things using Autoencoders and Dynamic Thresholds," *Journal of Machine and Computing*, vol. 4, no. 1, pp. 1–10, 2024, doi: 10.53759/7669/jmc202404001.

[34] M. H. Abidi, S. Chintakindi, A. U. Rehman, and M. K. Mohammed, "Elucidation of Intelligent Classification Framework for Hydroponic Lettuce Deficiency Using Enhanced Optimization Strategy and Ensemble Multi-Dilated Adaptive Networks," *IEEE Access*, vol. 12, pp. 58406–58426, 2024, doi: 10.1109/ACCESS.2024.3392482.

[35] M. C. Imam, D. Trisnawarman, Hugeng, and H. K. T. Sari, "Big Data and Internet of Things Web Service Management to Support Salt Agriculture Automation," *International Journal of Advanced Computer Science and Applications*, vol. 13, no. 12, pp. 295–300, 2022, doi: 10.14569/IJACSA.2022.0131236.

[36] E. A. Al-Shahari, G. Aldehim, M. Aljebreen, J. S. Alqurni, A. S. Salama, and S. Abdelbagi, "Internet of Things Assisted Plant Disease Detection and Crop Management using Deep Learning for Sustainable Agriculture," *IEEE Access*, 2024, doi: 10.1109/ACCESS.2024.3397619.

[37] H. M. Oliveira *et al.*, "An autonomous Internet of Things spectral sensing system for in-situ optical monitoring of grape ripening: design, characterization, and operation," *Comput Electron Agric*, vol. 217, 2024, doi: 10.1016/j.compag.2023.108599.

[38] M. San Emeterio de la Parte, J.-F. Martínez-Ortega, P. Castillejo, and N. Lucas-Martínez, "Spatio-temporal semantic data management systems for IoT in agriculture 5.0: Challenges and future directions," *Internet of Things*, vol. 25, p. 101030, 2024, doi: https://doi.org/10.1016/j.iot.2023.101030.

[39] A. Alameen, "Improving the Accuracy of Multi-Valued Datasets in Agriculture Using Logistic Regression and LSTM-RNN Method," *TEM Journal*, vol. 11, no. 1, pp. 454–462, 2022, doi: 10.18421/TEM111-58.

[40] E. M. Kenny, E. Ruelle, M. T. Keane, and L. Shalloo, "A Hybrid Model that Combines Machine Learning and Mechanistic Models for Useful Grass Growth Prediction," *Comput Electron Agric*, vol. 219, 2024, doi: 10.1016/j.compag.2024.108805.

[41] O. Elsherbiny, L. Zhou, Y. He, and Z. Qiu, "A novel hybrid deep network for diagnosing water status in wheat crop using IoT-based multimodal data," *Comput Electron Agric*, vol. 203, 2022, doi: 10.1016/j.compag.2022.107453.

[42] E. Anastasiou, A. T. Balafoutis, and S. Fountas, "Applications of extended reality (XR) in agriculture, livestock farming, and aquaculture: A review," *Smart Agricultural Technology*, vol. 3, p. 100105, 2023, doi: https://doi.org/10.1016/j.atech.2022.100105.

[43] G. Büyüközkan and D. Uztürk, "Integrated design framework for smart agriculture: Bridging the gap between digitalization and sustainability," *J Clean Prod*, vol. 449, p. 141572, Apr. 2024, doi: 10.1016/J.JCLEPRO.2024.141572.

[44] R. Svs and R. Mukkamala, "A CNN Bidirectional LSTM framework for predicting monsoon rainfall in India," *Acta Hydrologica Slovaca*, vol. 24, no. 2, pp. 214–220, 2023, doi: 10.31577/ahs-2023-0024.02.0024.

[45] S. Yalçın, M. Eşit, and M. İ. Yüce, "Climatological parameters estimation based on artificial intelligence techniques with particle swarm optimization and deep neural networks," *Journal of Universal Computer Science*, vol. 28, no. 10, pp. 1108–1133, 2022, doi: 10.3897/jucs.82370.

[46] Q. Li, Z. Li, W. Shangguan, X. Wang, L. Li, and F. Yu, "Improving soil moisture prediction using a novel encoder-decoder model with residual learning," *Comput Electron Agric*, vol. 195, 2022, doi: 10.1016/j.compag.2022.106816.

[47] N. Sakeef *et al.*, "Machine learning classification of plant genotypes grown under different light conditions through the integration of multi-scale time-series data," *Comput Struct Biotechnol J*, vol. 21, pp. 3183–3195, 2023, doi: 10.1016/j.csbj.2023.05.005.

[48] M. Billah, M. N. Adnan, M. R. Akhond, R. R. Ema, M. A. Hossain, and S. Md. Galib, "Rainfall prediction system for Bangladesh using long short-term memory," *Open Computer Science*, vol. 12, no. 1, pp. 323–331, 2022, doi: 10.1515/comp-2022-0254.

[49] C. Xiao, N. Chen, C. Hu, K. Wang, J. Gong, and Z. Chen, "Short and mid-term sea surface temperature prediction using time-series satellite data and LSTM-AdaBoost combination approach," *Remote Sens Environ*, vol. 233, 2019, doi: 10.1016/j.rse.2019.111358.

[50] A. A. Khan *et al.*, "Context Aware Evapotranspiration (ETs) for Saline Soils Reclamation," *IEEE Access*, vol. 10, pp. 110050–110063, 2022, doi: 10.1109/ACCESS.2022.3206009.

Authorized licensed use limited to: UNIVERSITAS GADJAH MADA. Downloaded on January 22,2025 at 06:13:41 UTC from IEEE Xplore. Restrictions apply.